Classical Field Theory — Exercise Sheet 28

Tensor variation, gravitational waves and boundary action

Hints and answers

Use Lecture 28's curvature, inverse-metric and action conventions. All metrics, variations and coordinate maps are smooth. Repeated spacetime indices run from zero to three. Write $\eta = \mathrm{diag}(1,-1,-1,-1)$ only for the flat background. None of the problems asserts global existence for Einstein evolution.

Exercise 28.1 (The full Palatini tensor calculation). Let g_s be a smooth family of Lorentz metrics in fixed coordinates and put

$$k^{\mu\nu} = \partial_s g_s^{\mu\nu}\big|_0, \qquad h_{\mu\nu} = \partial_s (g_s)_{\mu\nu}\big|_0 = -g_{\mu\alpha}g_{\nu\beta}k^{\alpha\beta}, \qquad C^\rho_{\mu\nu} = \delta\Gamma^\rho_{\mu\nu}.$$

Here ∇ is the background Levi–Civita connection.

(a) Prove that C is a tensor of type $(1,2)$ and derive its expression in terms of ∇h. Do not assume that connection coefficients themselves are tensor components.

(b) Vary the full coordinate Riemann formula. Expand both covariant derivatives of C, including all its index terms, to prove $\delta R^\rho{}_{\sigma\mu\nu} = \nabla_\mu C^\rho_{\nu\sigma} - \nabla_\nu C^\rho_{\mu\sigma}$. Contract to obtain the Palatini identity without setting $\Gamma = 0$.

(c) Contract again to derive the inverse-metric boundary vector $v^\mu = \nabla^\mu(g_{\alpha\beta}k^{\alpha\beta}) - \nabla_\nu k^{\mu\nu}$. This is the complete tensor argument promised by Lecture 28.

Hint. The difference of two connections is linear over smooth functions in both vector arguments. Differentiate metric compatibility to obtain C. The two terms from the lower derivative index cancel in the antisymmetrised covariant derivative because the background connection is torsion-free.

Answer. The identities are

$$C^\rho_{\mu\nu} = \frac{1}{2}g^{\rho\lambda}(\nabla_\mu h_{\lambda\nu} + \nabla_\nu h_{\lambda\mu} - \nabla_\lambda h_{\mu\nu}),$$
$$\delta\,\mathrm{Ric}_{\mu\nu} = \nabla_\rho C^\rho_{\nu\mu} - \nabla_\nu C^\rho_{\rho\mu}, \qquad g^{\mu\nu}\delta\,\mathrm{Ric}_{\mu\nu} = \nabla_\mu v^\mu.$$

The connection-product variations are essential away from normal coordinates.

Exercise 28.2 (Vacuum waves, gauge and the two polarisations). Set $g_{\mu\nu} = \eta_{\mu\nu} + s h_{\mu\nu} + O(s^2)$. Raise perturbation indices with η, write $h = \eta^{\mu\nu}h_{\mu\nu}$ and define $\bar h_{\mu\nu} = h_{\mu\nu} - \eta_{\mu\nu}h/2$.

(a) Derive $\Gamma^{(1)}$, $\mathrm{Ric}^{(1)}$ and $G^{(1)}$. Show explicitly that $G^{(1)}_{00}$ and $G^{(1)}_{0i}$ contain no second time derivatives of the perturbation. These are the linearized vacuum constraint equations on a slice.

(b) Under the covariant-metric Lie variation $h_{\mu\nu} \mapsto h_{\mu\nu} + \partial_\mu\xi_\nu + \partial_\nu\xi_\mu$, derive the trace-reversed change and its divergence. In harmonic gauge $\partial^\mu\bar h_{\mu\nu} = 0$, derive the vacuum wave equation. No analytic theorem establishing that gauge globally is assumed or requested.

(c) For a mode $h_{\mu\nu} = \epsilon_{\mu\nu}e^{-i\nu t + i\nu x^3}$ with $\nu > 0$, impose harmonic gauge. Use residual null-wave gauge parameters to set the four temporal amplitudes to zero and derive the complete transverse traceless matrix. Count the independent polarisations. The real field is obtained by taking real parts; the plane mode itself is not claimed to have finite total energy on $\mathbb{R}^3$.

Hint. At first order only derivatives of $\Gamma^{(1)}$ enter the curvature. For the constraints expand the time-index components before imposing harmonic gauge. For the mode use the covector $\ell_\mu = (\nu, 0, 0, -\nu)$ and a parameter $\xi_\mu = \alpha_\mu e^{-i\ell_\rho x^\rho}$.

Answer. In harmonic gauge $G^{(1)}_{\mu\nu} = -\Box\bar{h}_{\mu\nu}/2$. Residual null-wave gauge leaves

$$\epsilon^{\mathrm{TT}}_{\mu\nu} = \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & h_+ & h_\times & 0 \\ 0 & h_\times & -h_+ & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}.$$

There are two independent transverse traceless polarisations. The constraint components contain spatial and first time derivatives only.

Exercise 28.3 (The non-null boundary action). Let Σ be a smooth non-null boundary patch of a region D. All variations have compact support in a collar of the interior of this patch and vanish near its edges and every other boundary piece. A Gaussian collar for the background may be used as a geometric given:

$$g = \epsilon_n\, \mathrm{d}n^2 + \gamma_{ab}(n, y)\mathrm{d}y^a\mathrm{d}y^b, \qquad \Sigma = \{n = 0\}, \quad D \text{ locally has } n \le 0, \quad \epsilon_n = \pm 1.$$

The outward unit normal is $N = \partial_n$. Write $\mathrm{vol}_\Sigma = \iota_N \mathrm{vol}_g$ with outward-first orientation, $K = \nabla_\mu N^\mu$, and fix only the induced metric: $h_{ab}|_\Sigma = 0$ for $h_{\mu\nu} = \delta g_{\mu\nu}$. Normal components of h are initially unrestricted.

Gaussian normal coordinates for a smooth non-null hypersurface are the local construction described in Wald, *General Relativity*, §3.3, pp. 42–43. The boundary-action discussion is Appendix E.1, pp. 458–459. Use the explicit conventions in this question, including ϵ_n.

(a) Construct a vector field ξ vanishing on Σ such that $\widetilde{h} = h + \mathcal{L}_\xi g$ has $\widetilde{h}_{nn} = \widetilde{h}_{na} = 0$ near the patch. Derive and solve the required first-order normal equations. Explain why the induced variation is still zero.

(b) In this representative compute the normal flux of the Palatini vector and the variation of $2\epsilon_n \int_\Sigma K \mathrm{vol}_\Sigma$. Prove their cancellation.

(c) Use diffeomorphism invariance and Lecture 28's Bianchi identity to extend that result to the original arbitrary normal components. State the resulting gravitational action with the frozen overall sign. Specify why this is not a null-boundary or corner/joint formula.

Hint. The two normal Lie-variation components are $2\epsilon_n\partial_n\xi^n$ and $\gamma_{ab}\partial_n\xi^b + \epsilon_n\partial_a\xi^n$. For a Gaussian-preserving variation with $h_{ab} = 0$ on the boundary, $K = \gamma^{ab}\partial_n\gamma_{ab}/2$. The bulk Einstein pairing is unchanged by a boundary-fixing Lie variation.

Answer. The boundary-corrected gravitational action in this scope is

$$S_{\mathrm{grav}} = -\frac{1}{16\pi G_{\mathrm{N}}}\left[\int_D R_{\mathrm{sc}}\mathrm{vol}_g + 2\epsilon_n\int_\Sigma K\mathrm{vol}_\Sigma\right].$$

The Palatini flux is $-\epsilon_n\gamma^{ab}\partial_n\widetilde{h}_{ab}\mathrm{vol}_\Sigma$, cancelled by the boundary term's variation. No null boundary or joint term is included.